

Lars ten Hacken

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Education

Eindhoven University of Technology, MSc in Applied Physics Sept 2024 – July 2026

- GPA: 8.9/10.0, Cum Laude (highest honor). Selected Coursework: Advanced Fluid Dynamics, Soft Matter, Advanced CFD (LBM, Molecular Dynamics, Spectral Methods, FVM), Chaos, Gas Dynamics, Mathematical Physics.

Eindhoven University of Technology, Honors BSc in Applied Physics Sept 2020 – July 2023

- 215/180 ECTS, Honors Academy programme, hockey at E.S.H.V Don Quishoot, active member E.S.C (social/professional), president Helios (fraternal), several committees at study association and UniPartners Eindhoven.

Research Experience

Thesis Researcher, Eindhoven University of Technology – The Netherlands Nov 2025 – July 2026

- Mixed-convection phase-change systems (Prof. Federico Toschi), developed LBM code in C and OpenACC for $O(10^8)$ -gridpoint simulations on NVIDIA H100 GPUs.

Visiting Researcher, Tsinghua University – Beijing, China Feb 2026 – May 2026

- Experimental study on ice-water interface morphology in mixed convection (Prof. Chao Sun).

Visiting Researcher, University of California, Berkeley – CA, USA May 2025 – Nov 2025

- Phase-resolved ocean wave reconstruction and prediction from sparse swarm data (Prof. Reza Alam).

Output & Funding

Preparing: L.A. ten Hacken, Z. Wang, A. Gabbana, F. Toschi, “Effective viscosity in a sheared melt layer”.

Preparing: L.A. ten Hacken, F. Mohaghegh, R. Alam, “Spatiotemporal Reconstruction of Ocean Wave Fields from Sparse Measurements”.

APS Division of Fluid Dynamics Annual Meeting: Interact Session, Orlando, USA, Nov 2026.

J.M. Burgers Symposium: Poster and Pitch, The Netherlands, May 2026.

Funding: APS-DFD Enabling Grant (\$1,000); NL Scholarship Grant 25/26 & 24/25 (€2,500); TU/e FIE (€500).

Professional and Leadership Experience

R&D Scientist, FABBS (full-time) July 2026 – present

- Physics-based battery modelling for BMS development using Porous Electrode Theory and Doyle-Fuller-Newman models. Startup has two patents pending.

Columnist, Cursor Oct 2025 – July 2026

- Wrote opinion articles for TU Eindhoven’s independent news medium.

Workshop Leader and Speaker, De AI Workshop (8 h/week) Jan 2024 – July 2026

- Led hands-on workshops and public talks on artificial intelligence and its applications.

President, Junior Enterprise Netherlands (full-time) Sept 2023 – Aug 2024

- Led the national organisation comprising 12 local boards, 300+ students and €800k annual turnover across 250+ projects. Part of JE Europe.

Delegate, Netherlands Asia Honors Summer School – Hong Kong / Vietnam 2024

- Selected among 70 students from Dutch universities for an Asia-focused summer school and consulting project.

Treasurer, Business Core Eindhoven (16 h/week) 2022 – 2023

- Managed a € 60k budget and performed general organisational tasks for Eindhoven’s largest student symposium.

Skills

Python, C, JAX, Matlab, Wolfram Mathematica, PyTorch, TensorFlow, Linux, Git, Bash, OpenACC, Slurm. English (fluent), Dutch (native).